

Modeling Life Expectancy Using Health and Economic Indicators

Project Report

1. Introduction

Life expectancy is a key indicator of population health and socio-economic development, reflecting the combined influence of various health, demographic, and economic factors. Understanding the relationship between these factors and life expectancy can provide valuable insights into the broader determinants of population health.

This project examines the relationship between life expectancy and selected health and economic indicators using the **Life Expectancy WHO Updated dataset**. A regression-based framework is employed to model life expectancy and investigate the predictive relationships among the variables. The analysis includes OLS, Ridge, Lasso, and Principal Component Regression. The primary objectives of this project are to: load and inspect the data; perform a train/test split; fit an OLS baseline; check the major assumptions of linear regression; fit Ridge and Lasso regression; select the regularization parameter (alpha) for Ridge and Lasso via cross-validation on the training data; and compare all models on the same untouched test set.

No missing-value imputation is performed, since this version of the dataset is complete. All preprocessing (scaling, encoding) and hyperparameter tuning are fitted on the training data only, with the test set held out for final evaluation.

2. Dataset

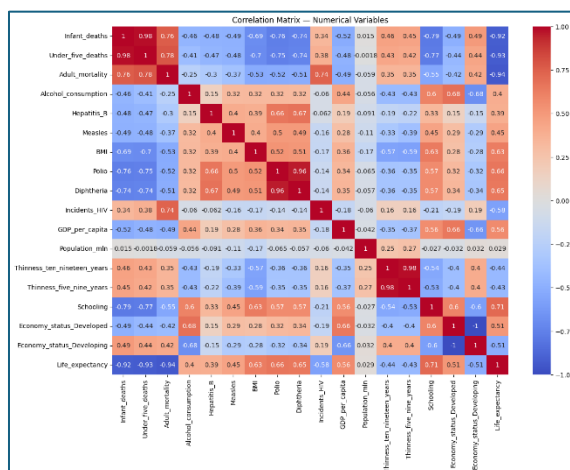
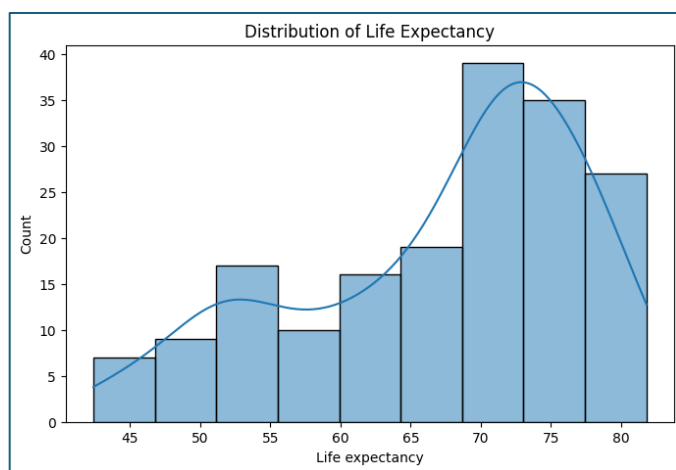
The source file, Life-Expectancy-Data-Updated.csv, contains 2,864 rows and 21 columns with no missing values. Its columns are: Country, Region, Year, Infant_deaths, Under_five_deaths, Adult_mortality, Alcohol_consumption, Hepatitis_B, Measles, BMI, Polio, Diphtheria, Incidents_HIV, GDP_per_capita, Population_mln, Thinness_ten_nineteen_years, Thinness_five_nine_years, Schooling, Economy_status_Developed, Economy_status_Developing, and the target, Life_expectancy.

We have restricted the analysis to a single cross-section of the panel data, filtering to the year 2003. This subset contains 179 countries across 9 regions.

Source: <https://www.kaggle.com/datasets/lashagoch/life-expectancy-who-updated>

2.1 Target Variable

The response variable is Life_expectancy. Its distribution is examined with a histogram/KDE plot, and a correlation heatmap of the numerical variables is produced as part of the exploratory analysis.



2.2 Predictor Selection and Feature Engineering

- Country is excluded from the model. Region is retained instead as the categorical geographic variable.
- Economy_status_Developed and Economy_status_Developing are collapsed into a single binary variable, Economy_status (1 = Developing, 0 = Developed), to avoid perfect linear dependence with the model intercept.
- Under_five_deaths is excluded due to direct overlap with Infant_deaths, which is retained as a predictor.
- Adult_mortality is excluded since it's a direct input into WHO's life table calculation.

The final predictor set used in every model is:

Numeric (14): Infant_deaths, Alcohol_consumption, Hepatitis_B, Measles, BMI, Polio, Diphtheria, Incidents_HIV, GDP_per_capita, Population_mln, Thinness_ten_nineteen_years, Thinness_five_nine_years, Schooling, Economy_status.

Categorical (1): Region (one-hot encoded with the first category dropped).

3. Methodology

3.1 Train/Test Split

The 179 observations from the 2003 cross-section are split 80/20 into training and test sets (`test_size = 0.20`, `random_state = 42`), producing 143 training observations and 36 test observations. The test set is held out completely and used only for the final model comparison to avoid data leakage.

3.2 Preprocessing Pipeline

A scikit-learn ColumnTransformer applies StandardScaler to the numeric features and OneHotEncoder (dropping the first category, handling unknown categories) to Region. This preprocessor is fit on the training data only and reused, unfitted-parameters-unchanged, to transform the test data — for every model in the notebook.

3.3 Ordinary Least Squares (OLS) Baseline

A LinearRegression model is fit on the preprocessed training features and evaluated on the untouched test set as the baseline for comparison with the regularized and dimensionality-reduced models.

3.4 OLS Assumption Checks

Six diagnostics are examined on the OLS model's training residuals:

- Linearity — residuals plotted against fitted values.
- Normality of residuals — histogram, Q-Q plot, and the Shapiro-Wilk test (capped at a random sample of 5,000 residuals).
- Homoskedasticity — residuals-vs-fitted plot and the Breusch-Pagan test.
- Independence — the Durbin-Watson statistic.
- Multicollinearity — variance inflation factors (VIF) computed on the transformed training design matrix.
- Influential observations — Cook's distance, computed from an equivalent statsmodels OLS fit, flagged against a $4/n$ reference threshold.

3.5 Principal Component Regression (PCR)

PCA is fit strictly after the train/test split and only on the training design matrix, retaining the number of components that explain 95% of training variance. Linear regression model is then fitted on the retained principal components, and

the fitted PCA transform is applied to the test set for evaluation. This avoids data leakage and tests whether dimensionality reduction preserves predictive performance.

3.6 Ridge and Lasso Regression

Ridge and Lasso are each wrapped in the same preprocessing pipeline as the OLS model. The regularization parameter α is selected via GridSearchCV over 100 candidate values spanning spaced evenly on a logarithmic scale, starting from 10^{-4} and ending at 10^4 using 3-fold cross-validation RMSE as the scoring metric. The Lasso model uses `max_iter = 100000` to ensure convergence.

3.7 Model Comparison

OLS, Ridge, Lasso, and PCR are all evaluated on the identical, previously untouched test set using R^2 , RMSE, and MAE, with lower RMSE/MAE and higher R^2 indicating better performance.

4. Results

4.1 OLS Baseline Performance

Metric	Value
Test R^2	0.9253
Test RMSE	2.6654
Test MAE	2.0682

On the training residuals, the mean is effectively zero ($-3.73e-15$) with a standard deviation of 2.276, consistent with a well-specified OLS fit.

4.2 Assumption Check Results

Check	Statistic	Conclusion
Normality (Shapiro-Wilk)	stat = 0.9816, p = 0.0511	Do not reject normality at the 5% level
Homoskedasticity (Breusch-Pagan)	LM = 30.52, p = 0.1063	No significant evidence of heteroskedasticity
Independence (Breusch-Godfrey)	LM = 3.1052, p = 0.0780	No significant evidence of residual autocorrelation at the 5% level
Influential points (Cook's distance)	max = 277.51	10 of 143 training observations exceed $4/n$

Linearity was assessed visually via the residuals-vs-fitted plot in the notebook and did not indicate a strong systematic pattern.

4.3 Multicollinearity (VIF)

VIF values were computed for every feature in the transformed training design matrix. Four features exceed the conventional $VIF > 10$ threshold for strong multicollinearity:

Feature	VIF
Thinness_five_nine_years	26.48
Thinness_ten_nineteen_years	24.18
Polio	15.24
Diphtheria	13.63

Economy_status (5.88) and Infant_deaths (5.85) fall in the 5–10 'potentially problematic' range; all remaining features, including every Region indicator, have VIF below 5. **This multicollinearity is the primary motivation for fitting Ridge and Lasso alongside OLS.**

4.4 Principal Component Regression

The 22 preprocessed (scaled/encoded) features are reduced to 11 principal components, which together explain 96.02% of the training variance (against a 95% target). The first principal component alone explains 42.15% of training variance.

Metric	Value
Test R ²	0.8798
Test RMSE	3.3804
Test MAE	2.8571

4.5 Ridge Regression

Metric	Value
Best alpha (3-fold CV)	3.3516
Test R ²	0.9260
Test RMSE	2.6533
Test MAE	2.1618

4.6 Lasso Regression

Metric	Value
Best alpha (3-fold CV)	0.5214
Test R ²	0.9212
Test RMSE	2.7377
Test MAE	2.1813

Of the 22 encoded coefficients, Lasso shrinks 19 to exactly zero, retaining only 3 predictors: Infant_deaths, Incidents_HIV, and GDP_per_capita.

4.7 Final Model Comparison

All four models were evaluated on the same held-out 36-observation test set.

Model	Train R ²	Train Adjusted R ²	Test R ²	Test RMSE	Test MAE	Tuned alpha
Ridge	0.9469	0.9372	0.9260	2.6533	2.1618	3.3516
OLS	0.9494	0.9402	0.9253	2.6654	2.0682	—
Lasso	0.9263	0.9247	0.9212	2.7377	2.1813	0.5214
PCR	0.9023	0.8941	0.8798	3.3804	2.8571	— (11 PCs, 96.02% variance)

Ridge regression achieves the lowest test RMSE and the highest test R², making it the best-performing model by that criterion, though OLS is close behind and has the lowest MAE. PCR trails the other 3 models on every metric.

5. Discussion

The OLS baseline satisfies its core assumptions reasonably well on this dataset: residuals do not show strong evidence of non-normality, heteroskedasticity, or autocorrelation at the 5% significance level. The main concern is multicollinearity — the two thinness measures and the Polio/Diphtheria immunization pair both show VIF well above the conventional threshold of 10, meaning their individual OLS coefficients should be interpreted with caution even though the model's overall fit is strong.

Ridge regression achieves the best test-set RMSE and R² among the four models, consistent with it addressing the multicollinearity observed in the VIF analysis. Lasso achieves a more interpretable, sparse model — 3 predictors versus 14 numeric predictors plus nine region indicators — at a modest cost in test RMSE and MAE relative to OLS and Ridge. PCR, despite retaining components that explain 96% of training variance, underperforms all other models on every test metric, suggesting that the directions of maximum variance in the predictor space are not the directions most predictive of life expectancy.

6. Conclusion

This analysis examined country-level Life Expectancy using health, socioeconomic and demographic indicators for the year 2003. The regression models achieved strong predictive performance, with the best models explaining more than 92% of the variation in Life Expectancy on the held-out test set. The diagnostic analysis indicated substantial multicollinearity among several health-related variables, which motivated the comparison of OLS with Ridge and Lasso regression. Ridge produced a small improvement over OLS, suggesting that coefficient shrinkage can be useful when predictors contain overlapping information. Lasso provided a more sparse model through variable selection, although with a modest reduction in predictive accuracy. PCR, despite retaining 96% of the variance in the predictors, performed noticeably worse, illustrating that high variance explained by PCA does not necessarily correspond to high predictive information about the response.

From a real-world perspective, the results suggest that Life Expectancy is strongly associated with a combination of health, economic, demographic and regional characteristics. Such a model could be used as a decision-support tool to identify countries or populations whose observed characteristics are associated with lower expected Life Expectancy and to help prioritize areas for further investigation or resource planning. However, the analysis is based only on cross-sectional data from 2003, so the results should not be interpreted as causal relationships or as a representation of present-day conditions. The model identifies statistical associations within the selected predictors rather than demonstrating that changing a particular factor would directly increase Life Expectancy. Further analysis using multiple years of data, additional health and socioeconomic variables, and appropriate longitudinal methods would be needed for stronger policy or causal conclusions.